

Converting Base Ten Fractions to Binary

To convert a decimal fraction to binary:

1. Multiply the fraction by 2.
2. Place the digit to the left of the decimal point (the whole number part of the result, either 0 or 1) immediately to the right of the radix point in the binary number.
3. Take the fractional part of the result from Step 1 and multiply it by 2.
4. Place the digit to the left of the decimal point (the whole number part of the result, either 0 or 1) on the right side of the binary number (to the right of the digit from Step 2).
5. Repeat multiplying the fractional part of the result by two and placing the whole number part of the result on the right end of the binary number. Stop when the fractional part of the multiplication result is zero.

Note: A decimal fraction with a finite number of digits may require a binary fraction with an infinite number of digits. The conversion from decimal to binary is not necessarily exact. In this case, stop after some number of conversion steps.

Examples:

1) Convert 0.37 from decimal to binary

$$0.37 \times 2 = 0.74$$

$$0.74 \times 2 = 1.48$$

$$0.48 \times 2 = 0.96$$

$$0.96 \times 2 = 1.92$$

$$0.92 \times 2 = 1.84$$

$$0.84 \times 2 = 1.68 \dots \text{(not converging)}$$

0.37 is approximately 0.010111 in binary

2) Convert 0.625 from decimal to binary

$$0.625 \times 2 = 1.25$$

$$0.25 \times 2 = 0.5$$

$$0.5 \times 2 = 1.0$$

0.625 is exactly 0.101 in binary